

Popular Deep Learning Model: Human Pose Estimation.

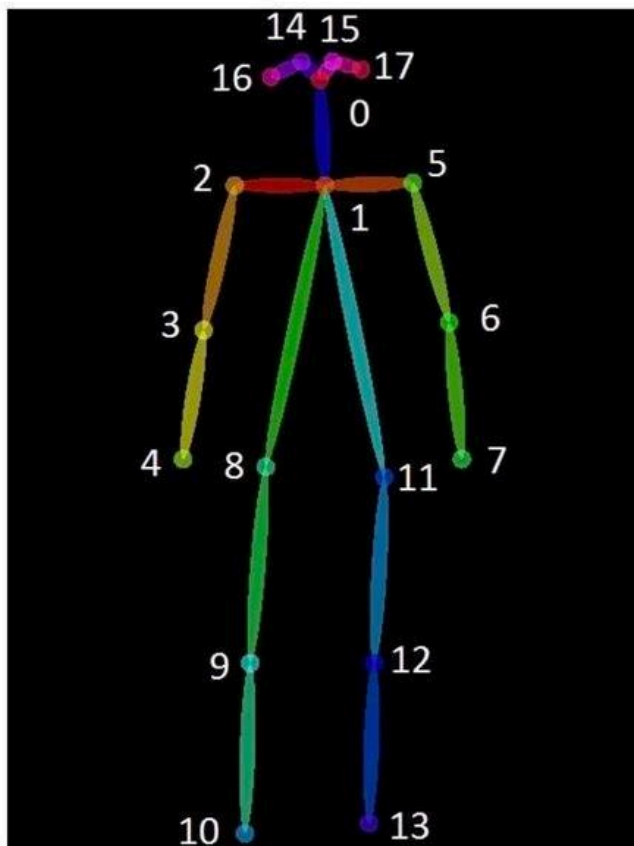
What is Human Pose Estimation?

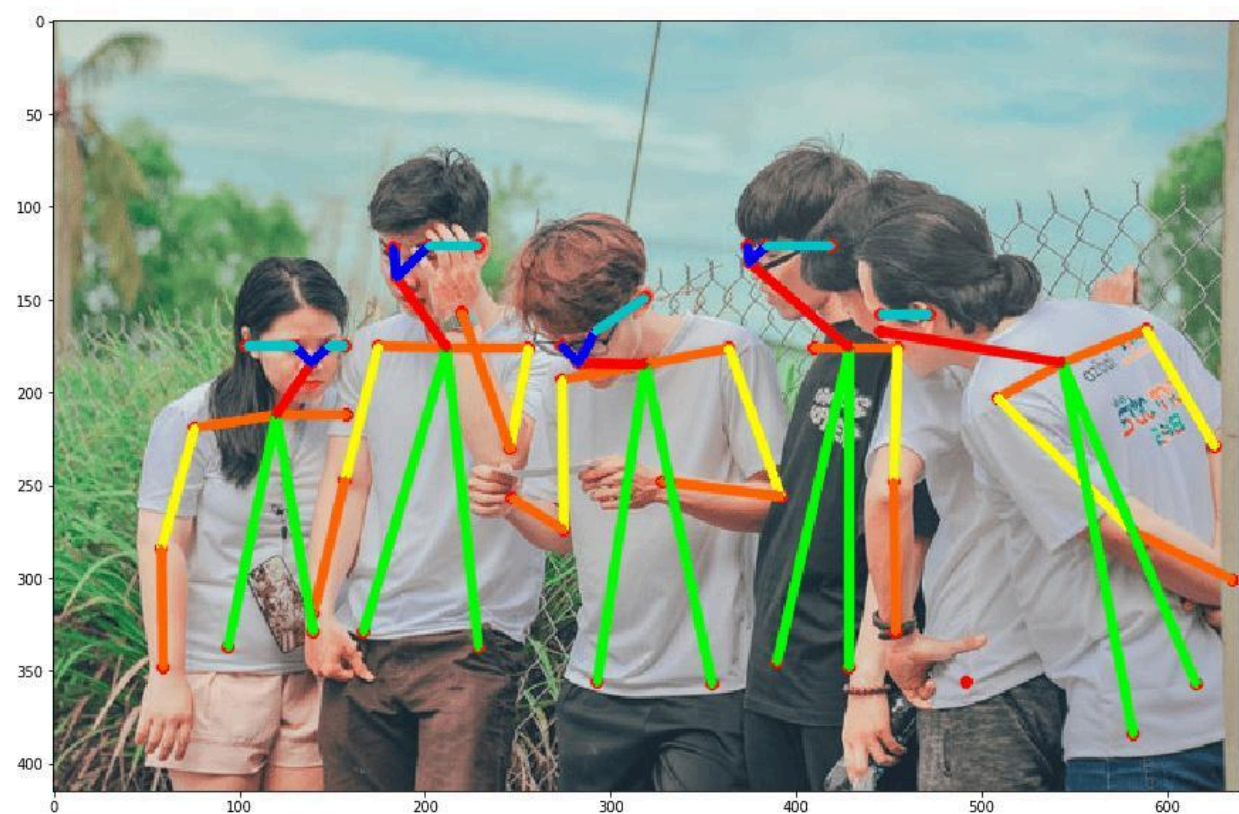
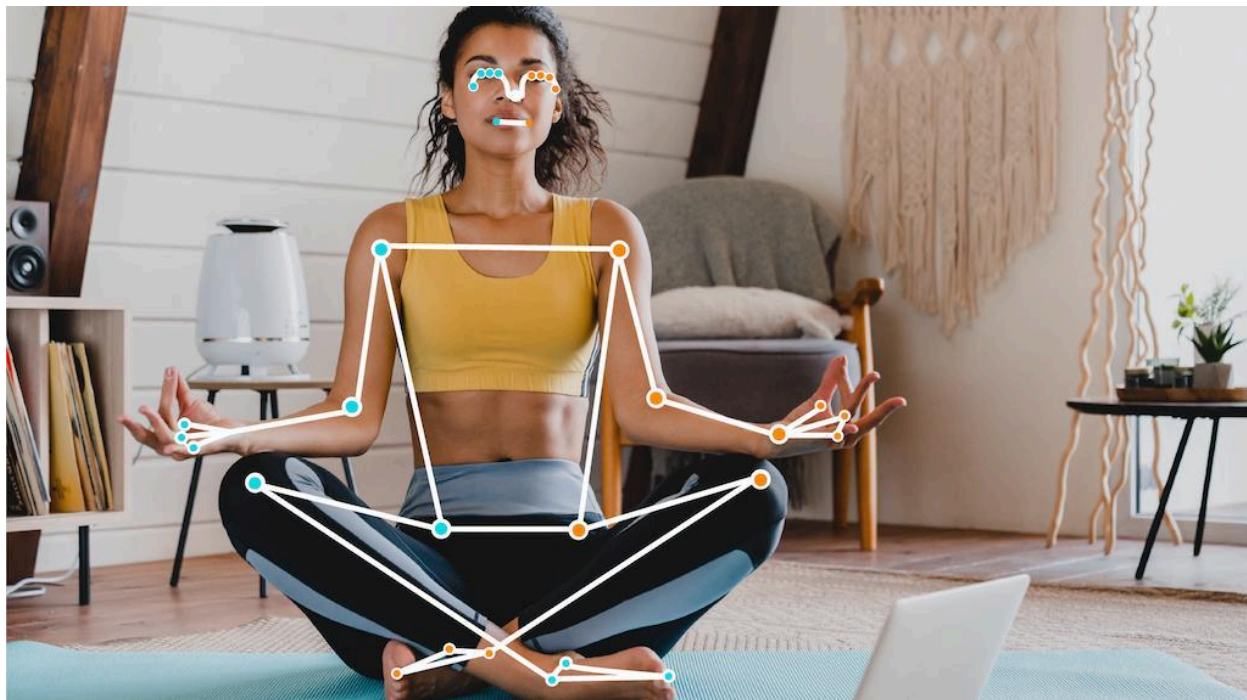
Human Pose Estimation is a Deep Learning technique that detects the **human body's key points (joints)** from an image or video.

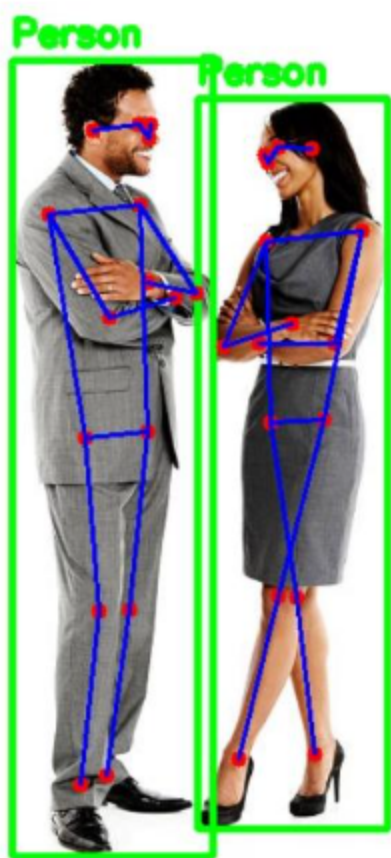
Examples of key points:

- Head
- Neck
- Shoulders
- Elbows
- Wrists
- Hips
- Knees
- Ankles

The model connects these points and creates a **skeleton structure**.







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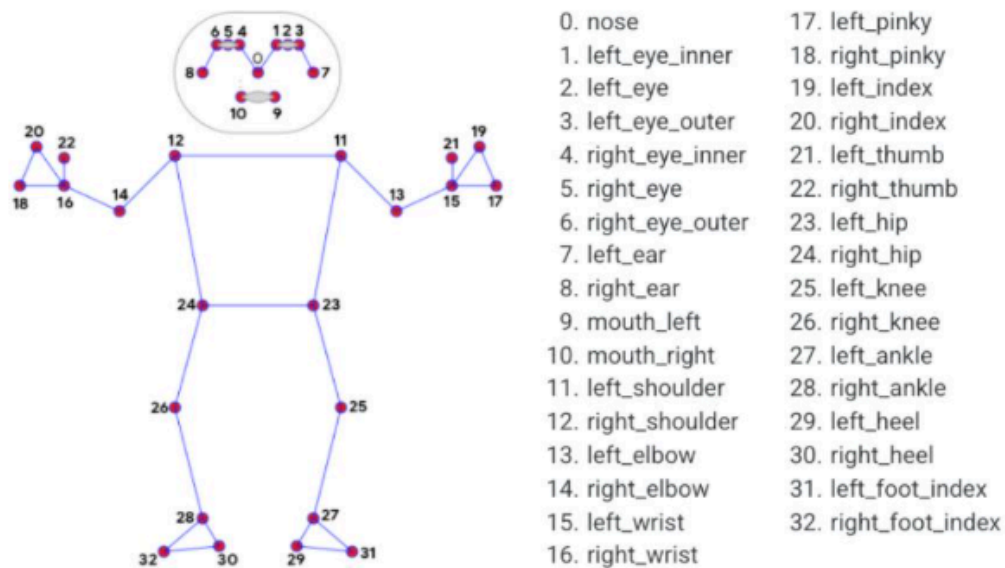


Figure 3: Human Pose Landmarks Detail

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What is shown in the slide?

Left Side

YOLOv7 Pose vs MediaPipe Pose

Both models detect:

- Person
- Body joints
- Skeleton lines

Output:

Camera Image



Pose Model



Body Key Points



Skeleton Drawing

Right Side (Top)

Different body postures are detected:

- Standing
- Walking
- Running
- Dancing
- Yoga

The model understands body movement by tracking joint positions.

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Right Side (Bottom)

The numbered points represent body landmarks.

Example:

Point	Body Part
0	Nose
11	Left Shoulder
12	Right Shoulder
13	Left Elbow
14	Right Elbow
15	Left Wrist
16	Right Wrist

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The exact numbering depends on the model.

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Real-Time Applications

Application	How Pose Estimation is Used
Fitness Trainer	Count exercises
Yoga Trainer	Check pose correctness
Dance Analysis	Track movements
Sports Analysis	Analyze player actions
Physiotherapy	Monitor rehabilitation
Gesture Control	Control computer using body movements
Fall Detection	Detect elderly falls
Smart Attendance	Detect human presence

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Example Project

AI Push-Up Counter

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Webcam
↓
MediaPipe Pose
↓
Detect Shoulder, Elbow, Wrist
↓
Calculate Arm Angle
↓
Count Push-Ups

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AI Yoga Trainer

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Webcam
↓
Pose Estimation Model
↓
Detect Body Joints

```

↓
Compare with Correct Yoga Pose
↓
Show "Correct" or "Incorrect"

Popular Pretrained Models

Example Project

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Popular Pretrained Models

Model	Use
MediaPipe Pose	Fast, beginner-friendly
YOLO Pose	Object + Pose detection
OpenPose	Research and detailed tracking
MoveNet	Lightweight and fast
BlazePose	Google's high-accuracy pose model

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Main Idea

Human Pose Estimation = Detect body joints + connect them as a skeleton + analyze movement.

This is why the slide shows stick-figure skeletons over people—the deep learning model is not just detecting a person; it is understanding the person's posture and movement.

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